

Aejandro De La Torre

LIS 440

wk11: Pandas Subsetting and Statistics

11/18/25

	A	B	C	D	E	F	G
1	Flavor	Sugar Cone	Cup	Waffle Cone	Cake Cone	Pretzel Cone	
2	Chocolate	45	38	27	25	33	
3	Strawberry	38	31	37	29	27	
4	Vanilla	28	38	27	44	42	
5	Moose Tracks	27	38	25	30	25	
6	Lemon	42	27	37	29	34	
7	Neapolitan	27	38	41	25	45	
8	Blue Moon	31	38	38	27	35	
9	Black Cherry	25	45	33	26	26	
10	Orange	28	38	29	29	43	
11	Cookie Dough	31	40	34	29	26	
12	Alejandro De	37	41	29	43	41	
13							
14	STATISTICS						
15	Largest:	45	45	41	44	45	
16	Smallest:	25	27	25	25	25	
17	Count:	11	11	11	11	11	
18	Total:	359	412	357	336	377	
19	Average:	32.64	37.45	32.45	30.55	34.27	
20	Median:	31	38	33	29	34	
21							
22							

Code

```
# Import the necessary packages
import pandas as pd
import os
from pathlib import Path

# Print version of pandas
print("Alejandro De La Torre's code....")
print(pd.__version__)

# -----
# File & Path Setup
# -----

# Set root directory (two levels up from this script)
root = Path(__file__).parent.parent.resolve()
```

```

# Show current working directory
print("Current Working Directory:", Path.cwd())

# Define key directories
data = root / 'data'
data_raw = data / 'raw'
data_processed = data / 'processed'

# Fetch the data
ic_df = pd.read_csv(data_processed / 'icecream_random.csv')
print("\n:Random Ice Cream data:")
print(ic_df)

# -----
# Sugar Cone Stats
# -----
print("\nSugar Cone Sales:")
sugar = ic_df["Sugar Cone"]
print(sugar)

print("\nTotal Sugar Cone Sales:")
print(sugar.sum())

print("\nLargest Sugar Cone Sale:")
print(sugar.max())

print("\nSmallest Sugar Cone Sale:")
print(sugar.min())

print("\nNumber of Sugar Cone Sales Data Points:")
print(sugar.count())

print("\nMedian Sugar Cone Sales for All Flavors:")
print(sugar.median())

print("\nAverage Sugar Cone Sales for All Flavors:")
print(sugar.mean())

# -----
# Cup Stats
# -----
print("\nCup Sales:")
cup = ic_df["Cup"]
print(cup)

print("\nTotal Cup Sales:")
print(cup.sum())

print("\nLargest Cup Sale:")
print(cup.max())

print("\nSmallest Cup Sale:")
print(cup.min())

```

```
print("\nNumber of Cup Sales Data Points:")
print(cup.count())

print("\nMedian Cup Sales for All Flavors:")
print(cup.median())

print("\nAverage Cup Sales for All Flavors:")
print(cup.mean())

# -----
# Waffle Cone Stats
# -----
print("\nWaffle Cone Sales:")
waffle = ic_df["Waffle Cone"]
print(waffle)

print("\nTotal Waffle Cone Sales:")
print(waffle.sum())

print("\nLargest Waffle Cone Sale:")
print(waffle.max())

print("\nSmallest Waffle Cone Sale:")
print(waffle.min())

print("\nNumber of Waffle Cone Sales Data Points:")
print(waffle.count())

print("\nMedian Waffle Cone Sales for All Flavors:")
print(waffle.median())

print("\nAverage Waffle Cone Sales for All Flavors:")
print(waffle.mean())

# -----
# Cake Cone Stats
# -----
print("\nCake Cone Sales:")
cake = ic_df["Cake Cone"]
print(cake)

print("\nTotal Cake Cone Sales:")
print(cake.sum())

print("\nLargest Cake Cone Sale:")
print(cake.max())

print("\nSmallest Cake Cone Sale:")
print(cake.min())

print("\nNumber of Cake Cone Sales Data Points:")
print(cake.count())
```

```

print("\nMedian Cake Cone Sales for All Flavors:")
print(ice_cream_sales.median())

print("\nAverage Cake Cone Sales for All Flavors:")
print(ice_cream_sales.mean())

# -----
# Pretzel Cone Stats
# -----
print("\nPretzel Cone Sales:")
pretzel = ic_df["Pretzel Cone"]
print(pretzel)

print("\nTotal Pretzel Cone Sales:")
print(pretzel.sum())

print("\nLargest Pretzel Cone Sale:")
print(pretzel.max())

print("\nSmallest Pretzel Cone Sale:")
print(pretzel.min())

print("\nNumber of Pretzel Cone Sales Data Points:")
print(pretzel.count())

print("\nMedian Pretzel Cone Sales for All Flavors:")
print(pretzel.median())

print("\nAverage Pretzel Cone Sales for All Flavors:")
print(pretzel.mean())

```

Shell

```

>>> %Run ic_stats.py
Alejandro De La Torre's code....
2.3.3
Current Working Directory: /Users/alexdelatorre/Library/CloudStorage/OneDrive-UW-Madison/Coursework/L I S
440/Lectures/Week 11/wk11 Pandas Subsetting and Statistics/code

:Random Ice Cream data:

```

	Flavor	Sugar Cone	Cake Cone	Pretzel Cone
0	Chocolate	45	25	33
1	Strawberry	38	29	27
2	Vanilla	28	44	42
3	Moose Tracks	27	30	25
4	Lemon	42	29	34
5	Neapolitan	27	25	45
6	Blue Moon	31	27	35
7	Black Cherry	25	26	26
8	Orange	28	29	43
9	Cookie Dough	31	29	26
10	Alejandro De La Torre's Favorite	37	43	41

```

[11 rows x 5 columns]

```

Sugar Cone Sales:

0	45
1	38
2	28
3	27
4	42
5	27
6	31
7	25
8	28
9	31
10	37

Name: Sugar Cone, dtype: int64

Total Sugar Cone Sales:

359

Largest Sugar Cone Sale:

45

Smallest Sugar Cone Sale:

25

Number of Sugar Cone Sales Data Points:

11

Median Sugar Cone Sales for All Flavors:

31.0

Average Sugar Cone Sales for All Flavors:

32.63636363636363

Cup Sales:

0	38
1	31
2	38
3	38
4	27
5	38
6	38
7	45
8	38
9	40
10	41

Name: Cup, dtype: int64

Total Cup Sales:

412

Largest Cup Sale:

45

Smallest Cup Sale:

27

Number of Cup Sales Data Points:

11

Median Cup Sales for All Flavors:

38.0

Average Cup Sales for All Flavors:

37.45454545454545

Waffle Cone Sales:

0 27

1 37

2 27

3 25

4 37

5 41

6 38

7 33

8 29

9 34

10 29

Name: Waffle Cone, dtype: int64

Total Waffle Cone Sales:

357

Largest Waffle Cone Sale:

41

Smallest Waffle Cone Sale:

25

Number of Waffle Cone Sales Data Points:

11

Median Waffle Cone Sales for All Flavors:

33.0

Average Waffle Cone Sales for All Flavors:

32.45454545454545

Cake Cone Sales:

0 25

1 29

2 44

3 30

4 29

5 25

6 27

7 26

8 29

9 29

10 43

Name: Cake Cone, dtype: int64

Total Cake Cone Sales:

336

Largest Cake Cone Sale:

44

Smallest Cake Cone Sale:

25

Number of Cake Cone Sales Data Points:

11

Median Cake Cone Sales for All Flavors:

29.0

Average Cake Cone Sales for All Flavors:

30.545454545454547

Pretzel Cone Sales:

0 33

1 27

2 42

3 25

4 34

5 45

6 35

7 26

8 43

9 26

10 41

Name: Pretzel Cone, dtype: int64

Total Pretzel Cone Sales:

377

Largest Pretzel Cone Sale:

45

Smallest Pretzel Cone Sale:

25

Number of Pretzel Cone Sales Data Points:

11

Median Pretzel Cone Sales for All Flavors:

34.0

Average Pretzel Cone Sales for All Flavors:

34.272727272727

>>>